BA2: Digital Korea

Week 9: Sentiment Analysis

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Today's Agenda

1. Week 8 review & check-in
2. What is sentiment analysis?
3. Dictionary-based approaches
4. Sentiment arithmetic: word + word + word = ?
5. Korean sentiment dictionaries
6. Application: Moon Jae-in's tweets
7. Demo: Sentiment analysis in Orange Data Mining
8. Looking ahead

Review & Check-In

Week 8 Review

Last week we covered:

- From sparse vectors (BoW) to dense vectors (embeddings)
- Word embedding concepts and nearest neighbors
- Semantic search and word similarity
- Document embeddings: representing entire documents as vectors

Key takeaway from Week 8

Embeddings capture **meaning**: words that appear in similar contexts end up near each other in vector space.

경제 − 성장 + 교육 ≈ 발전 (vector arithmetic)

Week 8 Review: Word Embeddings

The big idea: words as vectors capture semantic relationships.

한국 $-$ 서울 $+$ 도쿄 $\approx$ **일본**

Today we use a similar intuition, but instead of **meaning**, we measure **feeling**.

From meaning to sentiment

Word embeddings ask: *what does this word mean?*
Sentiment analysis asks: *how does this word feel?*

What Is Sentiment Analysis?

What Is Sentiment Analysis?

Sentiment analysis = determining whether a text expresses a positive, negative, or neutral attitude.

Examples in everyday life

- Product reviews (5 stars?)
- Social media posts (happy? angry?)
- News coverage (favorable? critical?)
- Political speeches (hopeful? alarming?)

Why it matters for research

- Track public opinion over time
- Compare tone across leaders
- Detect emotional framing
- Measure media bias

In computational text analysis

Sentiment analysis is a form of **measurement**: we assign a numeric value (valence) to text that captures its emotional direction.

Two Main Approaches

Dictionary-based (this week)

- Humans create word lists
- Each word gets a sentiment score
- Computer looks up and sums scores
- Transparent and interpretable
- No training data needed

Also called: lexicon-based, rule-based

Machine learning-based

- Train a model on labeled examples
- Model learns patterns from data
- Can capture context and nuance
- Requires training data
- Less transparent ("black box")

Also called: supervised classification

This week's focus

Dictionary-based methods — where the researcher defines the rules and the computer applies them at scale.

How Dictionary-Based Sentiment Works

1. Input text (tweet, speech, review)

↓

2. Tokenize into individual words

↓

3. Check each word against positive and negative word lists

↓

4. Count positive hits and negative hits

↓

5. Combine counts → document score

The logic: look words up in two lists, count the hits, combine.

A Simple Example

Input: 국민 여러분께 감사드리며 희망을 함께 만들어 갑시다
(Thank you, people of the nation — let's build hope together)

Kiwi content stems: 국민 감사 희망

Stem	**In KNU?**	**Contribution**
국민 (people)	not in dictionary	0
감사 (thanks)	positive list	$+1$
희망 (hope)	positive list	$+1$
Totals	2 **positive**, 0 **negative**	

Result

Two positive hits, zero negative hits, three content words. Orange normalises this to a percentage score (we'll see the exact formula shortly).

Sentiment Arithmetic

Sentiment Arithmetic: Word + Word + Word = ?

Last week: **word + word = meaning** (embedding analogies)
This week: **word + word + word = feeling** (sentiment scoring)

희망	+	경제	+	감사	=	Positive
$+1$		0		$+1$		$+2$
위기	+	경제	+	파괴	=	Negative
-1		0		-1		-2

Each positive word adds $+1$, each negative adds -1, words not in the dictionary are ignored.
The sum is the tweet's raw sentiment score.

Your Turn: Guess the Sentiment

What is the sentiment of each combination?

1. 행복 (happy) + 사랑 (love) + 희망 (hope) = **?**
2. 공포 (fear) + 고통 (pain) + 슬픔 (sadness) = **?**
3. 감사 (thanks) + 어려움 (difficulty) + 희망 (hope) = **?**
4. 기쁨 (joy) + 위기 (crisis) + 신뢰 (trust) = **?**

Think about it

Cases 3 and 4 are **mixed**. This is where dictionary methods get interesting — and where their limitations start to show.

Sentiment Arithmetic: Answers

#	Words	Hits	Result
1	행복 + 사랑 + 희망	3 pos, 0 neg	**Positive** $(+3)$
2	공포 + 고통 + 슬픔	0 pos, 3 neg	**Negative** (-3)
3	감사 + 어려움 + 희망	2 pos, 1 neg	**Positive** $(+1)$
4	기쁨 + 위기 + 신뢰	2 pos, 1 neg	**Positive** $(+1)$

Key insight

Dictionary methods treat every word independently. They don't understand *context*: "overcome a crisis" and "cause a crisis" would score the same for 위기.

When Arithmetic Breaks Down

Dictionary-based sentiment has known limitations

1. **Negation:** 행복하지 않다 (not happy) — "happy" is positive, but the sentence is negative
2. **Sarcasm:** 정말 잘하셨네요 (you did really well) — could be sincere or sarcastic
3. **Domain specificity:** 급등 (surge) — positive for stocks, negative for infections
4. **Mixed sentiment:** 어렵지만 희망적이다 (difficult but hopeful) — both negative and positive

So why use dictionaries?

Despite these limits, dictionary methods are **transparent**, **reproducible**, require **no training data**, and work well at scale when you understand their assumptions.

Korean Sentiment Dictionaries

Korean Sentiment Dictionaries: The Landscape

Several research dictionaries are publicly available for Korean

Dictionary	Size	Scale	Notes
KNU (Park et al. 2018)	14,854	-2 to $+2$	**Course dictionary (what we use)**
KOSAC (Jang et al. 2013)	∼11,275	5-class + intensity	SNU manual annotation on news
Chen & Skiena (2014)	44 langs	binary	Orange's built-in Multilingual option
NRC Emotion Lexicon	∼14,000	8 emotions +±	Cross-lingual, MT Korean
KSenticNet	5,465	continuous	Neural / concept-level

Why KNU?

Published Korean sentiment research standard. Simple word lists (positive, negative) that we load into Orange via Custom Dictionary. You can open the files in any text editor and read the words.

One Method, Two Places to See It

Same pipeline in Orange and in the interactive — numbers match

In Orange (you build this)

- File → Corpus
- Python Script (Kiwi)
- Corpus (re-map)
- Sentiment Analysis (Custom Dict)
- Box Plot

Five widgets, one script, loads KNU positive/negative files.

In the interactive (6 panels)

1. The Corpus — tweets per year
2. Dictionary Scoring — one tweet at a time
3. Score Distribution — histogram
4. By Period — box plot
5. Over Time — timeline + events
6. Explore Tweets — browse extremes

Same Kiwi + KNU pipeline, pre-computed.

Why both?

Orange gives you the tool. The interactive opens it up so you can see what the tool is doing to your tweets.

The KNU Korean Sentiment Lexicon

Our Week 9 dictionary — published research standard for Korean

Score	Meaning	Example	Translation
$+2$	Very positive	감사, 행복, 사랑	gratitude, happiness, love
$+1$	Positive	발전, 신뢰, 희망	development, trust, hope
0	Neutral	—	—
-1	Negative	위기, 가련한	crisis, pitiful
-2	Very negative	가난, 파괴, 화재	poverty, destruction, fire

How we use KNU in this course

Load `positive.txt` (words scored > 0) and `negative.txt` (words scored < 0) into Orange's Sentiment Analysis widget with the Custom Dictionary option. For this course we use polarity (positive vs. negative) rather than the full -2 to $+2$ intensity.

What's in the KNU Dictionary?

Content-word stems — same form Kiwi produces from tweets

Positive stems (samples)

- 감사 (gratitude) $\rightarrow +2$
- 행복 (happiness) $\rightarrow +2$
- 축하 (celebration) $\rightarrow +2$
- 사랑 (love) $\rightarrow +2$
- 희망 (hope) $\rightarrow +1$
- 발전 (development) $\rightarrow +1$

Negative stems (samples)

- 가난 (poverty) $\rightarrow -2$
- 파괴 (destruction) $\rightarrow -2$
- 화재 (fire disaster) $\rightarrow -2$
- 걱정 (worry) $\rightarrow -2$
- 어려움 (hardship) $\rightarrow -2$
- 위기 (crisis) $\rightarrow -1$

Files on the Data page

`positive.txt` and `negative.txt` — the word lists you load into Orange's Custom Dictionary. Full KNU with intensity scores in `SentiWord_Dict.txt` for R users.

Building Custom Dictionaries

Sometimes general dictionaries miss domain-specific sentiment

1. **Start with a general dictionary** (our positive/negative lists)
2. **Read a sample of your texts** — note domain-specific words
3. **Add domain terms:**
   - Political: 민주화 (democratization) → positive? 독재 (dictatorship) → negative?
   - Economic: 호황 (boom) → positive. 불황 (recession) → negative
4. **Validate:** check a sample of scored documents — do the scores make sense?

Remember GRS Chapter 2

Dictionary construction requires **domain knowledge**. You are making analytical choices about what counts as positive or negative in your research context.

Application: Moon Jae-in's Tweets

Case Study: Moon Jae-in on Twitter

3,148 tweets from @moonriver365 (2012–2020), scored with KNU

The corpus: @moonriver365

- 3,148 tweets (2012–2020), 9 columns
- Metadata: date, favorites, retweets
- Three periods: Pre-presidency (1,973), Transition (393), Presidency (782)

Moon Jae-in (문재인) — 19th President (2017–2022)

- Key topics: inter-Korean relations, COVID-19, Japan trade dispute, democracy
- Active Twitter presence, especially 2012 campaign (1,427 tweets)
- Tweeting declined sharply after taking office

Research questions:

1. Is the overall tone positive or negative?
2. Does sentiment change across the three periods?
3. Do different topics carry different sentiment?
4. How does sentiment relate to real-world events?

Sentiment Arithmetic with Real Tweets

A purely positive tweet: celebrating a birthday

Tweet (2018-01-24, presidency):
생일 축하, 고맙습니다. 생일을 챙기지 않는 삶을 살아왔는데 ⋯ 진심으로 감사드립니다.
(Thank you for the birthday wishes. I've lived without celebrating birthdays … I am truly grateful.)

Matched word	In dictionary
축하 (celebration)	positive
감사 (gratitude)	positive

2 positive matches, 0 negative. Orange's formula: $100 \times (2 - 0)/14$ tokens $= +14.3$.

Sentiment Arithmetic with Real Tweets

A mixed-sentiment tweet: grief, solidarity, hope

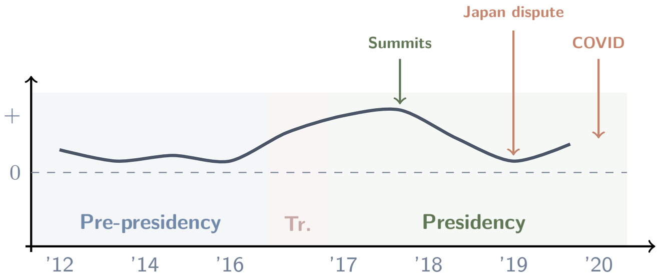
Tweet (2020-03-30, presidency):
어려움 속에서도 서로를 격려해가며 신뢰와 협력으로 재난을 이겨가고 있는 국민들께 한없는 존경과 감사를 드립니다.
(Even amid hardship, I offer deepest respect and gratitude to the people enduring this disaster with trust and cooperation.)

Matched word	**In dictionary**
신뢰 (trust)	**positive**
존경 (respect)	**positive**
감사 (gratitude)	**positive**
어려움 (hardship)	**negative**
재난 (disaster)	**negative**

3 positive, 2 negative in 20 tokens: $100 \times (3 - 2)/20 = +5.0$. The tweet leans positive because gratitude and respect outweigh the hardship words.

What Can Sentiment Over Time Reveal?

Our corpus spans three distinct political periods



- Does sentiment differ across the three periods? (Your assignment!)
- Do real-world events (summits, trade dispute, COVID) produce visible shifts?
- Political communication tends toward a positive baseline — does it hold?

Moment in Time: The Impeachment Vote

Tweet (2016-12-09, transition, 4,445 favorites):

국민이 이겼습니다. 대통령 탄핵은 끝이 아니라 시작입니다.

(The people have won. The presidential impeachment is not the end but the beginning.)

Dictionary scoring:

- Matched words: **0**
- 국민, 이기, 대통령, 탄핵, 시작 — none of these are in KNU
- Score: **0** (neutral)

Context:

- Posted day after the impeachment vote
- Tone is triumphant
- Dictionary says "neutral" — the method misses the political significance entirely

Moment in Time: COVID Zero

Tweet (2020-04-30, presidency, 25,952 favorites — most-liked COVID tweet):

72일 만의 코로나19 국내 확진자 0명. 총선 이후 14일간 선거로 인한 감염 0명. 대한민국의 힘, 국민의 힘입니다.

(For the first time in 72 days, zero domestic COVID-19 cases. Zero election-related infections in 14 days since the general election. This is the strength of Korea, the strength of the people.)

The dictionary problem

Words like 코로나, 확진자, and 감염 aren't in our general-purpose KNU dictionary at all (they're COVID-era vocabulary). Words like 힘 and 국민 are too short or too common to be scored. A human reader sees clear celebration; the dictionary sees 0 matches and scores it **neutral**.

Critical Reading: What the Dictionary Misses

Tweet (2019-08-02, presidency, 19,110 favorites):

우리는 다시는 일본에게 지지 않을 것입니다. 수많은 역경을 이겨내고 … 어려움을 극복할 역량이 있습니다.

(We will never again lose to Japan. We have overcome many adversities ... we have the capability to overcome difficulties.)

Dictionary scoring:

- 어려움 (difficulty) → **negative**
- 역경 (adversity) → **negative**
- 극복, 역량, 일본 — not in KNU
- Score: -16.67 (two negative hits in 12 content tokens)

Human reading:

- Tone is defiant and assertive
- Negative *toward Japan* specifically
- Positive *self-framing* (national strength)
- The dictionary sees just "difficulty"

Lesson

Dictionary-based sentiment measures **word-level valence**. It doesn't capture the **target** (negative toward whom?) or the **rhetorical purpose** (rallying, not despairing).

Orange Data Mining Demo

Preprocessing for Sentiment

Korean needs morphological tokenization — we use Kiwi

Why Kiwi for Korean?

- Korean is agglutinative: 행복합니다 (one eojeol) contains the stem 행복
- A whitespace tokenizer would treat 행복합니다 as one unsplittable token
- **Kiwi** is a standard Korean morphological analyzer — it extracts morphemes and POS tags

What our Python Script does

Tokenize each tweet with Kiwi, keep content-word morphemes (NNG, NNP, VA, VV) of length ≥ 2, join them with spaces, and store the result in a `processed_text` column. That's it — the standard Korean preprocessing pipeline.

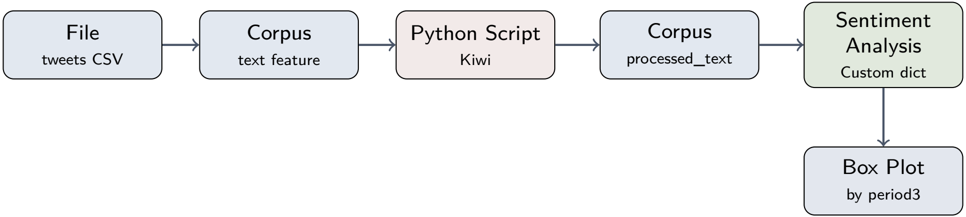
Demo: Sentiment Analysis in Orange

Kiwi preprocessing + Custom Dictionary with KNU word lists

1. **Load the data**
   - **File** → select `moon_twitter.csv`
   - **Corpus** → text feature: `text`
2. **Preprocess with Kiwi**
   - **Python Script** → paste `sentiment_preprocessing_*.py` from the Data page
   - Adds a `processed_text` column with NNG/NNP/VA/VV content tokens
3. **Re-map text feature**
   - Second **Corpus** widget → set "Used text features" to `processed_text`
4. **Score sentiment**
   - **Sentiment Analysis** → Method: **Custom Dictionary**
   - Load `positive.txt` and `negative.txt` (KNU word lists, on Data page)
5. **Explore**
   - **Box Plot** → subgroup by `period3`

Orange Workflow Overview

Five widgets: tokenize, match, count



What each widget does

File / Corpus: load tweets, mark `text` as document body. **Python Script**: Kiwi tokenizer, keeps content words (NNG/NNP/VA/VV). **Second Corpus**: remaps text feature to `processed_text`. **Sentiment Analysis** (Custom Dict): matches tokens against `positive.txt` / `negative.txt`. **Box Plot**: compare sentiment across `period3`.

Using the Sentiment Analysis Widget

Custom Dictionary lets you load your own word lists

Settings to choose:

1. **Method** $\rightarrow$ "Custom Dictionary"
2. Click the file browser button and load `positive.txt`
3. Click again and load `negative.txt`
4. Connect the output to a **Box Plot**

What's happening inside

The widget tokenizes each document (already done by Kiwi in your Python Script), counts unique matches against each word list, and computes $\texttt{sentiment} = 100 \times (\text{pos hits} - \text{neg hits})/\text{num tokens}$. A positive score means more positive matches than negative, normalized by document length.

Interpreting Your Results

Once you have sentiment scores, ask:

1. **Distribution:** What is the overall shape? Mostly positive? Skewed?
2. **Outliers:** Which documents scored extremely high or low? Do they make sense when you read them?
3. **Patterns:** Does sentiment correlate with time, topic, or another variable?
4. **Validation:** Read 10–20 documents across the score range. Does the algorithm's judgment match yours?

Always validate!

Grimmer, Roberts & Stewart: "All quantitative models of language are wrong — but some are useful" (Ch. 2). Validation is not optional.

Looking Ahead

For Next Week

Week 10: Topic Modeling (LDA)

- Latent Dirichlet Allocation (LDA)
- Discovering hidden thematic structure in a corpus
- Choosing the number of topics
- Interpreting and labeling topics

Recommended reading:

- Grimmer, Roberts & Stewart — Chapter 13: Topic Models

For Next Week

Interactive exercise, assignment, and preparation

Interactive exercise:

- Explore the **Sentiment Analysis: Moon Jae-in's Tweets** interactive on the course website
- Step through all 6 panels to see exactly what Orange is doing with your tweets

Assignment:

- Build the 5-widget Orange workflow on `moon_twitter.csv`: **File** → **Corpus** → **Python Script** (Kiwi) → **Corpus** → **Sentiment Analysis** (Custom Dict with `positive.txt`, `negative.txt`) → **Box Plot**
- Write 2–3 sentences: which period is most positive, and which matched words from the interactive explain why?

Preparation:

- Download `moon_twitter.csv`, `sentiment_preprocessing_*.py`, `positive.txt`, `negative.txt` from the **Data & Scripts** page